

Hongxiang (David) Qiu

Assistant Professor
Department of Epidemiology and Biostatistics
Michigan State University
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Current position

Michigan State University

Assistant Professor, Epidemiology and Biostatistics 2023–now
Research areas: causal inference, survival analysis, semiparametric inference, machine learning

Education

University of Washington

PhD, Biostatistics 2016–2021
Dissertation title: *In pursuit of automated statistical inference under minimal assumptions using machine learning tools*
Dissertation advisors: Marco Carone, PhD and Alex Luedtke, PhD

The Chinese University of Hong Kong

B.Sc., Mathematics; Minor in Statistics 2012–2016

Previous Work Experience

Wharton School, University of Pennsylvania

Postdoctoral researcher, Statistics 2021–2023
Supervisor: Eric Tchetgen Tchetgen, PhD and Edgar Dobriban, PhD
Research areas: machine learning, causal inference, semiparametric inference

Kaiser Permanente Washington Health Research Institute

Graduate Research Assistant 2016–2021
PRimary care Opioid Use Disorders Treatment (PROUD) Trial
Advisor: Jennifer F. Bobb, PhD

Awards and Fellowships

University of Washington

Scholarship for 6th Seattle Symposium in Biostatistics October 2020
2018 Donovan J. Thompson Award October 2018

Chinese University of Hong Kong

Dean's List, 2015–2016	July 2016
College Head's List, 2014–2015	January 2016
Department/Programme Scholarship, 2014–2015	January 2016
Undergraduate Mathematics Scheme	November 2015
Shaw College Global Exposure Award Scheme 2014–15	April 2015
Heung To Educational Fund Mathematics Scholarship	April 2015
Reaching Out Award	April 2015
1978 Mathematics Alumnus Li Sze-lim Scholarships	January 2015
College Head's List, 2013–2014	January 2015
Department/Programme Scholarship, 2013–2014	January 2015
CUHK Golden Jubilee Scholarship for Outgoing Exchange Student, 2014–2015	September 2015
Yasumoto International Exchange Scholarships	July 2014
Dean's List, 2013–2014	July, 2014
Undergraduate Mathematics Scheme	June 2014
Dragon Crowd "SCHIESSER" Scheme	April 2014
College Head's List, 2012–2013	January 2014
Department/Programme Scholarship, 2012–2013	January 2014
Outstanding Essay Award, 2012–2013	November 2013
Dean's List, 2012–2013	July 2013

Publications

(*: mentee; †: equal contribution.)

Methodology

1. **Qiu H** (2024). Non-Plug-In Estimators Could Outperform Plug-In Estimators: a Cautionary Note and a Diagnosis. *Epidemiologic Methods*, 13(1), 1–11. <https://doi.org/10.1515/EM-2024-0008>
2. **Qiu H**, Shi X, Miao W, Dobriban E, Tchetgen Tchetgen E (2024). Doubly robust proximal synthetic controls. *Biometrics*, 80(2). <https://doi.org/10.1093/biomtc/ujae055>.
3. **Qiu, H**, Tchetgen Tchetgen E, Dobriban E (2024). Efficient and multiply robust risk estimation under general forms of dataset shift. *The Annals of Statistics*, 52(4), 1796–1824.
4. **Qiu H**, Cook A, Bobb J (2024). Evaluating tests for cluster-randomized trials with few clusters under generalized linear mixed models with covariate adjustment: a simulation study. *Statistics in Medicine*, 43(2), 201–215. doi: 10.1002/sim.9950.
5. **Qiu H**, Dobriban E, Tchetgen Tchetgen E (2022). Prediction Sets Adaptive to Unknown Covariate Shift. *arXiv preprint: arXiv:2203.06126*. Accepted by *Journal of the Royal Statistical Society, Series B (Statistical Methodology)*.
6. **Qiu H**, Luedtke A (2022). Adversarial Meta-Learning of Gamma-Minimax Estimators That Leverage Prior Knowledge. *Electronic Journal of Statistics*, 17(2), 1996–2043.
7. **Qiu H**, Carone M, Luedtke A (2022). Individualized treatment rules under stochastic treatment cost constraints. *Journal of Causal Inference*, 10(1), 480–493.
8. **Qiu H**, Luedtke A, Carone M (2021). Universal sieve-based strategies for efficient estimation using machine learning tools. *Bernoulli*, 27(4), 2300–2336.

9. **Qiu H**, Carone M, Sadikova E, Petukhova M, Kessler R, Luedtke A (2020). Optimal individualized decision rules using instrumental variable methods. *Journal of the American Statistical Association* (with discussion), 116(533), 174–191.
10. Bobb J, **Qiu H**, Matthews, A, McCormack J, Bradley K (2020). Addressing identification bias in the design and analysis of cluster-randomized pragmatic trials: a case study. *Trials*, 21(1), 289.
11. **Qiu H**, Luedtke A, van der Laan M (2019). Contribution to discussion of “Entropy Learning for Dynamic Treatment Regimes” by Jiang B, Song R, Li J, Zeng D. *Statistica Sinica*, 29(4): 1666–1678.

Application

12. Al-Sahab B, Kerver JM, Alshaarawy O, Bohnert KM, Elliott MR, **Qiu H**, Jaber A, Neelam H, & Paneth N (2025). Prevalence and characteristics of prenatal cannabis use in Michigan, USA: A statewide population-based pregnancy cohort. *Addiction*. 121(1), 126—137.
13. Wartko PD, Bobb JF, Boudreau DM, Matthews AG, McCormack J, Lee AK, **Qiu H**, Yu O, Hyun N, Idu AE, Campbell CI, Saxon AJ, Liu DS, Altschuler A, Samet JH, Labelle CT, Zare-Mehrjerdi M, Stotts AL, Braciszewski JM, Murphy MT, Dryden D, Arnsten JH, Cunningham CO, Horigian VE, Szapocznik J, Glass JE, Caldeiro RM, Phillips RC, Shea M, Bart G, Schwartz RP, McNeely J, Liebschutz JM, Tsui JJ, Merrill JO, Lapham GT, Addis M, Bradley KA, PROUD Trial Collaborators (2023). Nurse Care Management for Opioid Use Disorder Treatment: The PROUD Cluster Randomized Clinical Trial. *JAMA Internal Medicine*. DOI: 10.1001/jamainternmed.2023.5701.
14. Wartko PD, **Qiu H**, Idu A, Yu O, McCormack J, Matthews AG, Bobb JF, Saxon AJ, Campbell CI, Liu D, Braciszewski JM, Murphy SM, Burganowski RP, Murphy MT, Horigian VE, Hamilton LK, Lee AK, Boudreau DM, Bradley KA (2022). Baseline representativeness of patients in clinics enrolled in the PRimary care Opioid Use Disorders treatment (PROUD) trial: comparison of trial and non-trial clinics in the same health systems. *BMC Health Services Research*, 22(1), 1–11.
15. Fitts W, Tassiou NR, Cisse FA, Vogel A, Atakla HG, Sakadi F, **Qiu H**, Conde ML, Balde AT, Bah AK, Hamani ABD, Anand P, Patenaude B, Mateen F (2019). School Status and its Associations among Children with Epilepsy in the Republic of Guinea. *Epilepsy & Behavior*, 97, 275–281.
16. Jang M, Sakadi F, Tassiou NR, Abass CF, Grundy SJ, Woga A, Kenda BA, Lamine CM, Talibé BA, **Qiu H**, Cohen JM, Carone M, Mateen FJ (2018). Impact of Poorly Controlled Epilepsy in the Republic of Guinea. *Seizure*, 61, 71–77.
17. Zhou Y[†], **Qiu H**[†], Xu S (2017). Modeling continuous admixture using admixture-induced linkage disequilibrium. *Scientific Reports*, 7, 1–10.

Preprint/under review

18. **Qiu H** (2025). Model-Agnostic Bounds for Augmented Inverse Probability Weighted Estimators’ Wald-Confidence Interval Coverage in Randomized Controlled Trials. *arXiv preprint: arXiv:2512.18898*.
19. **Qiu H**, Carone M, Luedtke A, & Gilbert PB. (2025). Multiply Robust Estimation of Conditional Survival Probability with Time-Varying Covariates. *arXiv preprint arXiv:2510.10372*.
20. Si W*, **Qiu H** (2025). Training-Set Conditionally Valid Prediction Sets with Right-Censored Data. *arxiv preprint: arXiv:2505.02213*.

Presentations

(*: mentee; †: equal contribution.)

Presenters

1. “Berry-Esseen type bound for Structure-Agnostic Semiparametric Treatment Effect Inference in Randomized trials.” **Qiu H.**
 - 2025 Joint Statistical Meeting, Nashville, TN (invited)
 - 2025 American Causal Inference Conference, Detroit, MI (poster)
2. “Novel Estimators of Minimally Important Change for Patient-Reported Outcomes.” **Qiu H.**
 - 2025 Research Day, Department of Epidemiology and Biostatistics, College of Human Medicine, Michigan State University, East Lansing, MI (invited)
3. “Efficient and Multiply Robust Risk Estimation under General Forms of Dataset Shift.” **Qiu H**, Tchetgen Tchetgen E, Dobriban E.
 - 2024 Joint Statistical Meeting, Portland, OR (invited)
4. “Doubly Robust Proximal Synthetic Controls.” **Qiu H**, Xu S, Miao W, Dobriban E, Tchetgen Tchetgen E.
 - 2023 American Causal Inference Conference, Austin, TX
5. “Prediction Sets Adaptive to Unknown Covariate Shift.” **Qiu H**, Dobriban E, Tchetgen Tchetgen E.
 - 2024 5th International Conference on Artificial Intelligence in Electronics Engineering, Bangkok, Thailand (invited)
 - 2022 Joint Statistical Meeting, Washington, D.C.
 - 2022 American Causal Inference Conference, Berkeley, CA (poster)
6. “Optimal individualized decision rules using instrumental variable methods.” **Qiu H**, Carone M, Sadikova E, Petukhova M, Kessler R, Luedtke A.
 - 2021 Joint Statistical Meetings, virtual (invited)
 - 2020 ENAR Spring Meeting, virtual
7. “Constructing asymptotically normal plug-in estimators with highly adaptive Lasso and data adaptive series.” **Qiu H**, Luedtke A & Carone M.
 - 2019 WNAR/IMS/JR Annual Meeting, Portland, OR

Contribution to presentations

8. “Robust Estimation of Stochastic Intervention Effects under Multistage Missingness.” Song S*, **Qiu H**, Chen H, Parks C & Luo Z. American Causal Inference Conference, Salt Lake City, UT, 2026
9. “Prevalence and characteristics of prenatal cannabis use in Michigan: Results from a statewide population-based pregnancy cohort.” Al-Sahab B, Alshaarawy O, Bohnert K, Jaber A, **Qiu H**, Kerver J & Paneth N. 37th Annual Meeting of the Society for Pediatric and Perinatal Epidemiologic Research (SPER), Austin, TX, 2024.

10. “Prevalence and characteristics of prenatal cannabis use in Michigan: Results from a statewide population-based pregnancy cohort.” Al-Sahab B, Alshaarawy O, Bohnert K, Jaber A, **Qiu H**, Kerver J & Paneth N. 37th Annual Meeting Society for Pediatric and Perinatal Epidemiologic Research, Austin, TX, 2024.

11. “Statistical challenges in the design of a pragmatic trial of primary care-based treatment for opioid use disorders: the PROUD trial.” Bobb JF, **Qiu H**, Matthews A & Bradley K. 12th International Conference on Health Policy Statistics, Charleston, SC, 2018

Reviewer for Funding Agencies

- 2025: National Institutes of Health (NIH)
- 2024: UK Research and Innovation

Reviewer for International Journals and Conferences

- *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*
- *Journal of the American Statistical Association*
- *Biometrics*
- *Journal of Causal Inference*
- *Biostatistics*
- *Statistica Sinica*
- *Journal of Computational and Graphical Statistics*
- *Electronic Journal of Statistics*
- *International Journal of Biostatistics*
- *Statistical Methods in Medical Research*
- *Stat*
- *Statistical Science*
- 5th International Conference on Artificial Intelligence in Electronics Engineering (AIEE 2024)
- 37th Conference on Uncertainty in Artificial Intelligence (UAI 2021)

Mentoring Experience

Gu Gong	2024–now
PhD student, Department of Epidemiology & Biostatistics, Michigan State University	
Shengfang Song	2024–now
PhD student, Department of Epidemiology & Biostatistics, Michigan State University	

Xi Gong	2024–now
Department of Statistics & PhD student, Probability, Michigan State University	
Wenwen Si	2024–2026
Department of Computer & PhD student, Information Science, University of Pennsylvania	

Teaching Experience

Michigan State University

EPI-826: Research Methods in Epidemiology	2025–now
<i>Instructor</i>	
Topics including: GLM, GEE, GLMM, sandwich robust SE, bootstrap, power calculation	
EPI-920: Inference under Semiparametric and Nonparametric Models (Independent Study)	Spring 2024
<i>Instructor</i>	
Topics including: (efficient) influence function, semiparametric efficiency bound, one-step correction, multiple robustness	
EPI-826: Research Methods in Epidemiology	Fall 2024
<i>Guest instructor</i>	
Topics including: generalized linear mixed model, generalized estimating equation	

University of Washington

STAT 583: Advanced Theory of Statistical Inference	Spring 2019
<i>Tutor</i>	
Topics including: empirical processes, semiparametric efficiency	
Instructors: Alex Luedtke, PhD and Marco Carone, PhD	
BIOST 557: Applied Statistics and Experimental Design	Winter 2019
<i>Teaching Assistant</i>	
Topics including: one/two-sample t-test, linear models, GLMs, causation versus correlation	
Instructor: Brian Lerous, PhD	
STAT 582: Advanced Theory of Statistical Inference	Winter 2018
<i>Grader</i>	
Topics including: Bayes methods, decision theory, UMPU test	
Instructor: Jon A. Wellner, PhD	

Software

CanonicalFamilyExtra	Extra canonical link family objects for generalized linear models. Particularly useful for fitting logistic or log-linear models for unbounded continuous outcomes.
RiskEstDShift	Efficient risk estimation under dataset shift.
APACpredset	Asymptotically Probably Approximately Correct prediction sets under unknown covariate shift.
CAMer	Continuous Admixture Modeling based on the result of iMAAPs.